

DREAM MORE TRANSPORT MANAGEMENT SYSTEM

**(Ride-Sharing and Smart Transport
Platform)**

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1.1 Introduction

Transportation plays a vital role in the economic and social development of modern societies. Efficient transportation systems enable people to travel conveniently, support business activities, and facilitate access to essential services such as education, healthcare, and employment. With the rapid growth of urban populations and increasing mobility demands, cities around the world are facing significant challenges in providing reliable, affordable, and efficient transportation services.

Recent advancements in information and communication technologies have transformed many sectors, including transportation. The emergence of digital ride-hailing platforms and intelligent transport systems has revolutionized the way people access transportation services by enabling real-time communication between passengers and drivers, route optimization, and electronic payment mechanisms. These technologies have contributed to improving convenience, reducing operational inefficiencies, and enhancing the overall user experience.

Despite these technological developments, many urban areas, particularly rapidly growing cities such as Addis Ababa, continue to experience transportation-related challenges. Traffic congestion, long waiting times, increasing travel costs, and inefficient utilization of available vehicles negatively affect both passengers and drivers. Moreover, existing transportation services often fail to provide mechanisms that allow individuals traveling along similar routes to share rides and split transportation expenses, resulting in underutilization of resources and higher transportation costs.

To address these challenges, there is a growing need for intelligent transportation solutions that leverage modern technologies such as geospatial data, Global Positioning System (GPS) tracking, real-time communication, and dynamic fare calculation. Such systems can improve transportation accessibility, enhance transparency, optimize resource utilization, and contribute to safer and more sustainable urban mobility.

Therefore, this project proposes the development of the Dream More Transport Management System, a web-based ride-hailing and transport management platform designed to connect passengers and drivers efficiently while providing real-time GPS tracking, distance-based fare calculation, ride scheduling, payment management, and opportunities for individuals traveling along common routes to share rides and transportation costs. The proposed system aims to improve transportation efficiency, reduce waiting times, provide transparent pricing, and enhance the overall safety and convenience of urban transportation services

1.2 Problem Statement

Urban transportation has become an increasing challenge in rapidly growing cities around the world. In cities such as Addis Ababa, the growing population and rising demand for mobility have placed significant pressure on existing transportation services.

Daily commuters, students, employees, drivers, and transport service providers are among the groups most affected by these challenges. The problem becomes particularly severe during peak hours, rush periods, and times of high demand when passengers experience long waiting times and difficulties finding available vehicles.

Although ride-hailing and taxi services have improved access to transportation, several gaps still remain. Many vehicles operate with unused seats while numerous passengers travel in similar directions independently. Existing systems provide limited support for connecting individuals traveling along common routes, preventing them from sharing rides and splitting transportation costs. In addition, passengers and drivers often lack transparency regarding route information and fare calculations.

These issues are commonly observed in densely populated urban areas and contribute to increased travel expenses, traffic congestion, fuel consumption, and inefficient utilization of transportation resources. Consequently, passengers spend more time waiting for transportation, drivers lose opportunities to maximize vehicle occupancy, and the overall efficiency of the transportation system is reduced.

Therefore, there is a need for a smart transport management system that enables real-time matching of passengers and drivers, allows individuals traveling along common routes to share rides and costs, and provides transparent route and fare information. Such a system would improve transportation efficiency, reduce waiting times and travel expenses, and promote safer and more sustainable urban mobility.

1.3 Objectives of the Project

1.3.1 General Objective

The general objective of this project is to develop a smart and efficient Dream More Transport Management System that enhances transportation accessibility by connecting passengers and drivers in real time, enabling ride sharing among individuals traveling along common routes, and providing transparent fare calculation and vehicle tracking services.

1.3.2 Specific Objectives

The specific objectives of the project are:

1. To design and develop a secure passenger and driver registration and authentication system.
2. To provide real-time ride booking and ride scheduling functionalities.
3. To implement GPS-based vehicle tracking and route optimization mechanisms.
4. To develop a dynamic distance-based fare calculation system that ensures transparent pricing.
5. To enable individuals traveling along common routes to share rides and divide transportation costs fairly.
6. To provide ride history management for both passengers and drivers.
7. To implement a notification system for ride requests, ride status updates, and important system events.
8. To develop a driver and passenger rating mechanism that improves trust, accountability, and service quality.
9. To integrate payment management functionalities for efficient transaction handling.
10. To provide an administrative dashboard for monitoring users, rides, payments, and overall system activities.
11. To improve transportation efficiency and reduce waiting times, particularly during periods of high demand, promote safer, more affordable, and sustainable urban transportation through efficient utilization of available vehicles.

1.4 Scope of the Project

The scope of the Dream More Transport Management System is limited to the development of a web-based platform that facilitates efficient interaction between passengers and drivers while improving transportation accessibility and management. The system focuses on providing essential ride-hailing and transport management functionalities that support real-time communication, route optimization, and transparent fare calculation.

The proposed system covers passenger management, which includes user registration, authentication, profile management, and access to ride-related services. It also supports driver management by allowing drivers to register, maintain their profiles, and manage ride requests.

The system provides ride management functionalities such as ride booking, ride scheduling, trip monitoring, and maintenance of ride history. Real-time GPS tracking and route optimization mechanisms are incorporated to enable efficient navigation and improve transportation efficiency. Dynamic fare calculation based on travel distance and route information is also included to ensure transparent pricing.

Furthermore, the system supports payment management, user notifications, and a rating mechanism that allows passengers and drivers to provide feedback after completing trips. Administrative functionalities are included through an admin dashboard that enables monitoring and management of users, rides, and system activities.

In addition, the system provides a mechanism that allows individuals traveling along common routes to share rides and transportation costs, thereby improving vehicle utilization and reducing travel expenses.

However, the project does not cover autonomous vehicle technologies, traffic signal management, vehicle maintenance management, integration with government transportation infrastructure, or advanced artificial intelligence-based traffic prediction systems. The system is also limited to transportation management and does not include logistics, parcel delivery services, or public transit scheduling.

1.5 Market Research & Competitor Analysis

1.5.1 Competitor Comparison Table

A side-by-side comparison of all five platforms across the key dimensions.

Category	Uber	Bolt	Lyft	Careem	inDrive	RIDE	Feres	Yango
Founded	2009, USA	2013, Estonia	2012, USA	2012, UAE	2012, Russia	2016, Ethiopia	2020, Ethiopia	2023 (Ethiopia entry); Yango Group HQ: Dubai
Coverage	70+ countries, 15,000+ cities	50+ countries, 600+ cities	US & Canada only (644+ cities)	12 countries, 100+ cities (MENA & South Asia)	48 countries, 1000+ cities	Addis Ababa and Djibouti	Addis Ababa and surrounding areas	Addis Ababa; expanding via franchise model
Active Users	202M active users monthly	200M+ customers	29.2M active riders	50M+ registered users	150M+ total downloads	3M+ customers, 70K+ drivers, 400+ corporate accounts	60K+ drivers	Growing since 2023 entry; partnered with G2G IT Group
Pricing Model	Algorithmic / surge pricing	Algorithmic / surge pricing	Algorithmic + Price Lock for commuters	Algorithmic / surge pricing	Passenger - proposed fare negotiation	Fixed upfront fare; no surge pricing	Fixed upfront fare; no surge pricing	Fixed upfront fare; no surge pricing
Surge Pricing	Yes	Yes	Yes (Price Lock partially mitigates)	Yes	No	No	No	No
Driver Commission	20 - 40% commission	~25%	Driver guaranteed 70% minimum	20 - 25%	5 - 8% (lowest in industry)	Not publicly disclosed	Lower than competitors; no commission during	Not publicly disclosed

							launch period	
Advance Booking	Up to 90 days	Up to 90 days	Up to 30 days	Yes	No	Yes	Yes	Not confirmed
Ride Types	UberX, Comfort, Black, WAV, Green, Assist	Standard, Premium, EV	Wait & Save, Standard, Comfort, XL, Black, WAV, Green, Silver	Go, Comfort, Executive, Max, Kids, Plus	Standard city rides Intercity	Sedan, Minivan, Minibus, Comfort, Comfort EV, English-speaking driver	Standard car (up to 6), Minivan, Minibus	Economy, comfort (A/C, newer vehicles)
GPS Tracking	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trip Sharing	Yes	Yes	Yes	Yes	Yes	Not confirmed	Not confirmed	Yes - sharable link for family and friends
SOS / Emergency Button	Yes	Yes (+ immediate welfare call)	Yes	Yes	Yes	Yes (SOS number)	Yes (immediate team response confirmed)	Yes - connects directly to emergency services
Phone Number Masking	Yes	Yes	Yes	Yes	Partial	Not confirmed	Not confirmed	Not confirmed
PIN Ride Verification	Yes	Yes	No	Yes	No	No	No	No
Two-Way Rating System	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Female Driver Preference	Yes	Yes	Yes (Women+ Connect)	No	No	No	No	No
Accessibility Rides (WAV)	Yes (selected markets)	No	Yes (Selected cities)	No	No	No	No	No

Child-Safe Ride Option	No	No	No	Yes (Careem Kids)	No	No	No	No
Ride Insurance	No	Yes	No	No	No	No	No	No
Route Preview Before Acceptance	No	No	No	No	Yes	No	No	No
Passenger Selects Driver	No	No	No	No	Yes	No	No	No
Cost Splitting	No	No	No	No	No	No	No	No
Food Delivery	Yes (Uber Eats)	Yes (Bolt Food)	No	Yes (Careem Food)	No	Yes (50+ restaurants)	No	No
Digital Wallet	No	No	No	Yes (Careem Pay)	No	No	No	No
Cash Payment	Yes (Selected Markets)	Yes (Selected Markets)	No	Yes	Yes	Yes	Yes	Yes
Loyalty / Membership	Yes (Uber One)	Yes (Bolt Plus)	Yes (Lyft Pink)	No	No	Not confirmed	Not confirmed	Not confirmed
Landmark-Based Navigation	No	No	No	Yes	No	Not confirmed	Not confirmed	Not confirmed
Biggest Strength	Global scale and comprehensive safety suite	Africa / Europe presence and ride insurance	Price Lock and driver transparency	Super app ecosystem and Careem Kids	Fare negotiation and zero surge pricing	Pioneer brand, largest fleet, inclusive payment, EV rides, maternal	Largest driver network, Feres Miles loyalty program,	Proprietary maps, strongest safety policy, immediate driver suspension on complaint

						transport CSR		
Biggest Weakness	Surge pricing and hidden fees	Driver cancel trip and app crashes	North America only; surge pricing persists	Billing fraud and captain cancel trip	Bait-and-switch pricing and poor GPS	Driver GPS failures, fare inflation via detours, unreliable refund process	Fare manipulation, app instability, driver accountability gaps, safety incidents	Map color design causes missed pickups, declining vehicle quality, driver earnings too low
Google Play Store Rating	~4.8 / 5	~4.1 / 5	~4.1 / 5	~4.2 / 5	~4.8 / 5	~4.7 / 5	~4.4 / 5	~4.8 / 5

1.5.2 Feature Gap Analysis

The following identifies critical features that are absent or poorly implemented across the five platforms researched, and explains how our system addresses each gap.

1.5.2.1 Fixed or negotiated fare with no surge pricing

Missing from: Uber, Bolt, Lyft, Careem

Passengers in our system confirm a fixed upfront price or negotiate a fare directly. No algorithmic price spikes occur under any condition.

1.5.2.2 Fare locked after agreement

Missing from: All five platforms (violated in practice by inDrive drivers)

Once a fare is agreed upon in our system, it is binding for both parties and cannot be changed mid-trip by either side.

1.5.2.3 Cost splitting between passengers

Missing from: All eight platforms

Passengers split the fare with travel companions directly in the app. Each party is notified, confirms their share, and is charged individually.

1.5.2.4 Passenger selects their own driver

Missing from: Uber, Bolt, Lyft, Careem, RIDE, Feres, Yango

Passengers view available drivers with ratings, vehicle details, and proposed fares before choosing who to ride with.

1.5.2.5 Route visible before acceptance

Missing from: Uber, Bolt, Lyft, Careem, RIDE, Feres, Yango

Both driver and passenger review the full route before the ride is confirmed.

1.5.2.6 Driver cancel penalty - no passenger charge

Missing from: Bolt, Careem, RIDE, Feres

If a driver cancels or manipulates a passenger into cancelling, the driver bears the penalty. Passengers are never charged for driver-initiated cancellations.

1.5.2.7 Landmark-based navigation

Missing from: Uber, Bolt, Lyft, inDrive, RIDE, Feres, Yango

Passengers set pickup and drop-off using local landmarks, not just GPS coordinates or formal street addresses.

1.5.2.8 Accurate, reliable GPS with clear map design

Missing from: inDrive, RIDE, Feres, Yango

Our system invests in GPS accuracy with landmark-based fallback for areas without reliable street addresses. Map UI designed with high contrast so drivers never miss pickup or drop-off points.

1.5.2.9 Phone number masking (both sides, no exceptions)

Missing from: inDrive (Partial), RIDE, Feres, Yango (not confirmed)

Both driver and passenger phone numbers are fully masked in all in-app communication with no exceptions.

1.5.2.10 Transparent charge history

Missing from: Careem, Feres

Every charge, adjustment, and refund is logged and visible to the passenger in their trip history in real time.

1.5.2.11 Human-led customer support with real resolution

Missing from: All eight platforms

Complaints are routes to a human support agent. Disputes are tracked with resolution timeline and the outcome is communicated to the user.

1.5.2.12 Simplified app mode for low-tech users

Missing from: Uber, Bolt, Careem, inDrive, Feres

An optional simplified interface with larger text and fewer steps, designed for users less familiar with smartphones.

1.5.2.13 Cash payment support

Missing from: Lyft

Cash is accepted alongside digital payments and wallet options.

1.5.2.14 Ride insurance coverage

Missing from: Uber, Lyft, Careem, inDrive, RIDE, Feres, Yango

All trips are covered by ride insurance for unexpected injuries or medical expenses during a ride.

1.5.2.15 Sustainable driver fare and commission structure

Missing from: Yango (fares too low for drivers), Uber (commission too high)

Commission rate and fare structure balanced so drivers can cover costs and earn fairly, without pricing passengers out.

1.5.2.16 Driver navigation training before onboarding

Missing from: RIDE, Feres, Yango

Drivers must demonstrate confident in-app map navigation before their first ride. Calling passengers for directions is not acceptable.

1.5.2.17 Fare protection against driver detours

Missing from: RIDE, Feres, inDrive

If a driver takes a longer or incorrect route, the passenger pays only the originally confirmed fare. Detour-based fare inflation is blocked by design.

1.5.3 What Our System Improves

1.5.3.1 Ride Sharing

No existing platform, local or global, offers a built-in way for multiple passengers traveling together to share a single ride booking with individual accountability. Our system allows passengers to book a shared ride together. Each rider confirms participation and is tracked under the same trip, reducing empty seats and lowering individual costs.

1.5.3.2 Cost Splitting

Riders who wish to split costs must settle offline using bank transfer or mobile money apps. No platform, local or global, handles this natively. Our system integrates cost splitting directly into the payment flow. Each passenger confirms their share and is charged individually, with cash, mobile payment, and digital payment options supported.

1.5.3.3 Pricing Transparency

Surge pricing across Uber, Bolt, Lyft, and Careem causes unpredictable fares that passengers describe as the most frustrating aspect of ride-hailing. Local platforms (RIDE and Feres) have reported fare inflation through detours and hidden additions. inDrive's negotiation model, while promising, is frequently violated by drivers who renegotiate after acceptance. Our system enforces fixed upfront fares confirmed before ride acceptance. No surge pricing exists. Once fare is agreed, it is locked and cannot be changed by either party for any reason. Detour-based fare inflation is blocked - if a driver goes the wrong way, the passenger pays the original fare only.

1.5.3.4 GPS Tracking

inDrive has widely reported GPS inaccuracy across multiple markets. Careem's app changes passenger pickup locations without user input. Uber has a long-standing bug that directs drivers to incorrect pickup points. Ride and Feres drivers regularly call passengers for directions. Yango's map color scheme causes drivers to miss pickup points. Our system addresses this with accurate real-time GPS and a landmark-based fallback for areas without reliable street addressing. Map UI designed with high contrast for clear driver guidance. Both parties confirm the pickup and drop-off points before the ride begins, eliminating location mismatches from the start.

1.6. Stakeholders

Stakeholders

The Dream More Transport Management System involves the following key stakeholders:

1. Passengers

Passengers are users who request rides, track drivers in real time, and pay for transportation services. They benefit from fast, reliable, and cost-effective mobility.

2. Drivers

Drivers provide transportation services by accepting ride requests, navigating routes, and completing trips. They earn income through the platform and depend on efficient ride matching.

3. Administrators

Administrators manage the entire system, including user accounts, driver verification, ride monitoring, payments, and system analytics.

4. System Owner / Company

The organization responsible for maintaining and scaling the platform, ensuring profitability, reliability, and continuous improvement.

5. Third-Party Services

External services such as payment gateways, mapping APIs, and notification services that support system functionality.

1.7. User Roles

User Roles

Passenger

- Register and log in to the system.
- Request and schedule rides.
- Track drivers in real time using GPS.
- Share rides with other passengers.
- View ride history and payment records.
- Rate drivers after completing trips.

Driver

- Register and get verified by admin.
- Accept or reject ride requests.
- Navigate to pickup and drop-off locations.
- Update ride status (started, completed, canceled).
- View earnings and ride history.

Administrator

- Manage passenger and driver accounts.
- Approve or reject driver registrations.
- Monitor active rides in real time.
- Generate system reports and analytics.
- Manage payments and disputes.

1.8. User Stories

User Stories

Passenger User Stories

- As a passenger, I want to register an account so that I can use the ride-hailing services.
- As a passenger, I want to log in securely so that I can access my personal dashboard.
- As a passenger, I want to request a ride so that I can travel to my destination easily.
- As a passenger, I want to view available nearby drivers so that I can get a quick ride.
- As a passenger, I want to track my driver in real time so that I know when they will arrive.
- As a passenger, I want to share my ride with others so that I can reduce travel costs.
- As a passenger, I want to see fare estimates before booking so that I can make informed decisions.
- As a passenger, I want to split the fare automatically when sharing rides so that payment is fair.
- As a passenger, I want to view my ride history so that I can track my past trips.
- As a passenger, I want to rate my driver so that I can provide feedback on service quality.

Driver User Stories

- As a driver, I want to register on the platform so that I can offer transportation services.
- As a driver, I want to log in securely so that I can access ride requests.
- As a driver, I want to receive ride requests so that I can accept or reject them.
- As a driver, I want to navigate to pickup and drop-off locations so that I can complete trips efficiently.
- As a driver, I want to update ride status so that passengers are informed in real time.
- As a driver, I want to view my earnings so that I can track my income.
- As a driver, I want to see my ride history so that I can analyze my performance.

- As a driver, I want to rate passengers so that I can maintain service quality.

Admin User Stories

- As an admin, I want to manage user accounts so that I can control system access.
- As an admin, I want to verify drivers so that only trusted drivers operate on the platform.
- As an admin, I want to monitor active rides so that I can ensure system safety.
- As an admin, I want to generate reports so that I can analyze system performance.
- As an admin, I want to manage complaints and disputes so that issues are resolved efficiently.
- As an admin, I want to view analytics dashboards so that I can make data-driven decisions.

System-Level User Stories

- As a system, I want to calculate fares based on distance so that pricing is accurate.
- As a system, I want to optimize routes so that travel time is reduced.
- As a system, I want to match passengers with nearby drivers so that ride requests are fulfilled quickly.
- As a system, I want to send notifications so that users stay updated on ride status.
- As a system, I want to process payments securely so that transactions are reliable.
- As a system, I want to support ride sharing so that multiple passengers can travel together.
- As a system, I want to operate at scale so that it can support thousands of users simultaneously.

1.9. Functional Requirements

Functional Requirements (SRS)

User Roles

The system has four primary users:

1. Passenger

A passenger is a registered user who can request rides, share rides, make payments, and rate drivers.

2. Driver

A driver is a registered and verified user who accepts ride requests, transports passengers, and receives payments.

3. Administrator

The administrator manages the entire platform, including users, drivers, rides, payments, reports, and system monitoring.

4. Guest User

A guest can browse the application and register but cannot book rides until an account is created.

Functional Requirements

FR-01 User Registration

The system shall allow passengers and drivers to register using their phone number or email address.

FR-02 User Authentication

The system shall allow registered users to securely log in and log out using email/phone number and password or OTP verification.

FR-03 Profile Management

The system shall allow users to view, edit, and update their personal information.

FR-04 Driver Verification

The system shall allow drivers to upload required documents (license, vehicle registration, insurance) for admin verification before becoming active.

FR-05 Ride Booking

The system shall allow passengers to request a ride by selecting pickup and destination locations.

FR-06 Ride Scheduling

The system shall allow passengers to schedule rides for future dates and times.

FR-07 Ride Sharing

The system shall allow passengers traveling in similar directions to share one ride, reducing travel costs and improving transportation efficiency.

FR-08 Driver Matching

The system shall automatically assign the nearest available driver based on GPS location and availability.

FR-09 GPS Tracking

The system shall display the real-time location of drivers and passengers throughout the trip.

FR-10 Route Optimization

The system shall calculate the shortest and fastest available route using GPS and mapping services.

FR-11 Fare Calculation

The system shall automatically calculate ride fares based on:

Distance

Travel time

Base fare

Traffic conditions

Ride-sharing discounts (if applicable)

FR-12 Cost Splitting

The system shall allow ride-sharing passengers to split transportation costs automatically according to predefined rules.

FR-13 Payment Management

The system shall support secure payment through:

Cash

Credit/Debit Cards

Mobile Money

Digital Wallets

FR-14 Ride Notifications

The system shall send real-time notifications for:

Ride request accepted

Driver arrival

Ride started

Ride completed

Payment confirmation

Ride cancellation

FR-15 Ride History

The system shall store previous rides including:

Pickup location

Destination

Driver

Fare

Payment status

Date and time

FR-16 Rating and Reviews

The system shall allow passengers and drivers to rate and review each other after completing a ride.

FR-17 Ride Cancellation

The system shall allow passengers and drivers to cancel rides according to the platform's cancellation policy.

FR-18 Emergency and Safety Features

The system shall provide an emergency SOS button allowing passengers and drivers to quickly notify emergency contacts or authorities during unsafe situations.

FR-19 Admin Dashboard

The system shall allow administrators to:

Manage passengers

Manage drivers

Verify driver accounts

Monitor rides

View reports

Handle complaints

Manage payments

FR-20 Reporting and Analytics

The system shall generate reports showing:

Total rides

Revenue

Active users

Driver performance

Ride-sharing statistics

Customer satisfaction ratings

User Stories

Passenger

US-01 As a passenger, I want to register an account so that I can book rides.

US-02 As a passenger, I want to request a ride so that I can travel to my destination.

US-03 As a passenger, I want to share my ride with other passengers so that transportation costs become cheaper.

US-04 As a passenger, I want to schedule rides in advance so that transportation is available when I need it.

US-05 As a passenger, I want to track my driver in real time so that I know when the driver will arrive.

US-06 As a passenger, I want to know the fare before confirming my ride so that pricing is transparent.

US-07 As a passenger, I want to pay using different payment methods so that payment is convenient.

US-08 As a passenger, I want to rate drivers after my trip so that service quality improves.

Driver

US-09 As a driver, I want to receive nearby ride requests so that I can earn income.

US-10 As a driver, I want to navigate using GPS so that I reach passengers efficiently.

US-11 As a driver, I want to receive payment confirmation immediately after completing rides.

US-12 As a driver, I want passengers to rate my service so that I build my reputation.

Administrator

US-13 As an administrator, I want to verify drivers before activation so that passengers remain safe.

US-14 As an administrator, I want to monitor rides in real time so that I can respond to issues quickly.

US-15 As an administrator, I want to generate reports and analytics so that I can make informed business decisions.

2. Non-Functional Requirements

The primary non-functional requirements for the Dream More Transport Management System prioritize reliable local-city availability, low-latency driver-passenger matching, and secure transaction handling, while remaining cost-effective and maintainable for deployment on standard cloud infrastructure.

Core Non-Functional Requirements (NFRs)

1. Availability and Reliability

1.Uptime Target: The core ride-booking and dispatch services must maintain 99.9% uptime (3 nines) during standard operational hours, with critical fallback modes (e.g., manual driver assignment) if the automated system fails.

2. Disaster Recovery: Database snapshots and configuration files must be backed up to a secondary region daily, enabling full system restoration within 2 hours in the event of catastrophic data loss.

3. Fault Tolerance: The system must gracefully handle failures of third-party mapping APIs (e.g., OpenStreetMap/Google Maps) by caching last-known route data and retrying with exponential backoff, preventing total service lockout.

2. Performance and Latency

1.Matching Speed: The automated dispatching algorithm must assign a nearby available driver to a passenger within under 5 seconds from the initial request, accounting for a city-wide pool of up to 500 active drivers.

2.Fare Calculation: Distance-based fare estimates, including dynamic surge multipliers, must be computed and displayed to the passenger within under 1 second after the destination is entered.

3.Real-Time Tracking: Driver GPS coordinates must refresh on the passenger's tracking screen every 5 to 10 seconds (adjustable based on network conditions) with a network latency of less than 500ms for position pings.

3. Scalability and Throughput

1.Geospatial Querying: The system must efficiently handle up to 1,000 concurrent driver location updates and 5,000 passenger search requests per minute, utilizing spatial indexing (e.g., PostGIS R-tree or MySQL geospatial indexes) to group drivers by city zone.

2. Concurrent Write Capacity: The database layer must support ingesting at least 100 concurrent GPS coordinate writes per second without degrading read performance for trip history queries.

3. Horizontal Scaling: The application backend (ride matching, payment, and notification services) must be stateless to allow for spinning up duplicate instances during local peak hours (e.g., festival events).

4. Security, Privacy, and Compliance

1. Data Anonymization: Driver and passenger phone numbers and precise pickup/drop-off coordinates must be automatically masked (pseudonymized) in historical trip logs and analytics databases to protect user privacy.

2. Payment Security: All financial transactions must be processed through a PCI-DSS Level 1 compliant third-party gateway (e.g., Stripe/PayPal), ensuring the system never persists raw credit card details locally.

3. Role-Based Access Control (RBAC): Admin dashboard access must be strictly tiered (e.g., Super Admin, Fleet Manager, Support Agent) using secure JWT tokens with enforced session timeouts after 30 minutes of inactivity.

5. Maintainability and Observability

1. Centralized Monitoring: All microservices (Ride Service, Driver Service, Notification Service) must expose structured health-check endpoints, with centralized logging via an ELK or Loki stack to isolate backend errors within 5 minutes of occurrence.

2. Feature Flagging: Engineering and operations teams must be able to enable or disable critical features (e.g., dynamic surge pricing, ride scheduling, or push notifications) remotely via configuration variables without requiring a full system redeployment.

3.Modular Testing: The core fare calculation and matching logic must be decoupled from the UI and GPS ingestion, allowing for offline unit testing with mock data to validate algorithmic changes before production rollout.

2.1. Use Cases

Use Cases

UC-01: Register User

Actor: Passenger / Driver

Flow:

1. User enters details.
2. System validates input.
3. Account is created

UC-02: Book Ride

Actor: Passenger

Flow:

1. Passenger enters destination.
2. System finds nearby drivers.
3. Ride request is sent.
4. Driver accepts ride.
5. Ride begins.

UC-03: Accept Ride

Actor: Driver

Flow:

1. Driver receives request.
2. Driver accepts or rejects.
3. If accepted, navigation starts.

UC-04: Track Ride

Actor: Passenger

Flow:

1. Passenger opens tracking screen.
2. System shows real-time driver location.

UC-05: Share Ride

Actor: Passenger

Flow:

1. Passenger selects ride sharing option.
2. System matches with nearby passengers.
3. Cost is split automatically.

UC-06: Process Payment

Actor: Passenger

Flow:

1. Ride ends.
2. System calculates fare.
3. Payment is processed securely.

UC-07: Admin Manage Users

Actor: Admin

Flow:

1. Admin accesses dashboard.
2. Views users.

3. Updates or removes accounts.

2.2. Constraints

Constraints

1. Technology Constraints

- The system must be built using **Next.js** for both frontend and backend implementation.
- **PostgreSQL** must be used as the primary relational database.
- **Redis** must be used for caching and performance optimization.
- External services such as **Cloudinary** must be used for media storage.

2. Performance Constraints

- The system must handle up to **10,000+ concurrent users** without significant performance degradation.
- API response time should remain under **2 seconds** under normal load conditions.

3. Network Constraints

- The system depends on stable internet connectivity for real-time GPS tracking and ride updates.
- Poor network conditions may affect location accuracy and ride synchronization.

4. Device Constraints

- The system must be accessible via modern web browsers (Chrome, Firefox, Edge).

- Mobile responsiveness is required, but a dedicated mobile app is not included in the current scope.

5. Third-Party Service Constraints

- Payment processing depends on external payment gateway APIs.
- Map and GPS services depend on third-party providers (e.g., Google Maps or equivalent).
- System functionality may be affected if external services experience downtime.

6. Security Constraints

- User authentication must be implemented using secure methods such as JWT.
- Passwords must be stored using strong hashing algorithms.
- All communication must be encrypted using HTTPS.

7. Development Constraints

- The project must be completed within the internship timeline.
- The system is developed by a small team (5 members), limiting parallel development capacity.
- Rapid prototyping is prioritized over perfect optimization in early phases.

8. Legal and Compliance Constraints

- User data must be handled according to basic data privacy principles.
- The system must not store sensitive payment details directly; it relies on third-party payment providers.

9. Scalability Constraints

- The system must be designed with horizontal scaling in mind.
- Stateless backend design is required to support load balancing.

2.3 Assumptions

Assumptions

1. User Behavior Assumptions

- Users (passengers and drivers) have basic knowledge of using web-based applications.
- Users will provide accurate personal and location information during registration.
- Drivers will actively update their availability status when using the system.

2. Connectivity Assumptions

- Users have access to a stable internet connection for booking rides and real-time tracking.
- GPS services will be available and accurate enough to support real-time location tracking.

3. Device and Platform Assumptions

- Users will access the system using modern web browsers such as Chrome, Firefox, or Edge.
- Devices used by users support JavaScript and modern web technologies.
- The system will initially be web-based (mobile application is not part of the current scope).

4. Third-Party Services Assumptions

- External services such as payment gateways, mapping APIs, and Cloudinary will be available and reliable.
- These third-party services will provide stable APIs for integration.

5. System Usage Assumptions

- The system is expected to operate under normal urban transportation conditions.
- Ride demand will vary but will remain within the designed scalability limits (up to 10,000+ concurrent users).
- Drivers will accept or reject ride requests in a timely manner.

6. Security Assumptions

- Users will keep their login credentials confidential.
- Authentication tokens (JWT) will be securely stored on the client side.

7. Operational Assumptions

- The system will be maintained and monitored by system administrators.
- Regular updates and maintenance will be performed to ensure system stability.